# DIP Exam 2024 solutions

1. (a) Let $f(x,y)$ denote an $N \times N$ −point two-dimensional sequence, that has zero value outside $0 \le x \le N-1,\ 0 \le y \le N-1$, where $N$ is an integer power of 2. In implementing the standard Walsh Transform of $f(x,y)$, we relate $f(x,y)$ to a new $N \times N$ −point sequence $W(u,v)$.

(i) Define the sequence $W(u,v)$ in terms of $f(x,y)$. Explain all the symbols used. **[1]**

**Answer**

$$W(u,v) = \frac{1}{N}\sum_{x=0}^{N-1}\sum_{y=0}^{N-1} f\,(x,y)(-1)^{\sum_{i=0}^{n-1}(b_i(x)b_{n-1-i}(u)+b_i(y)b_{n-1-i}(v))}$$

We use binary representation for the values of the independent variables $x$ and $u$. We need $n$ bits to represent them, $N = 2^n$. Hence, for the binary representation of $x$ and $u$ we can write:

$(x)_{10} = (b_{n-1}(x)b_{n-2}(x) \dots b_0(x))_2$

$(u)_{10} = (b_{n-1}(u)b_{n-2}(u) \dots b_0(u))_2$ with $b_i(x)$ 0 or 1 for $i = 0, \dots, n-1$.

(ii) For the case $N = 2$ and $f(x,y) = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$ calculate the Walsh transform coefficients. **[4]**

**Answer**

We know that $N = 2^n$ and therefore, in case of $N = 2$ we have $n = 1$, $x, y$ 0 or 1 and $b_0(0) = 0,\ b_0(1) = 1$. For $f(x,y) = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$ we calculate the Hadamard transform coefficients as follows.

$$H(u,v) = \frac{1}{N}\sum_{x=0}^{2-1}\sum_{y=0}^{2-1} f\,(x,y)\left[\prod_{i=0}^{n-1}(-1)^{(b_i(x)b_{n-1-i}(u)+b_i(y)b_{n-1-i}(v))}\right]$$

$$H(u,v) = \frac{1}{N}\sum_{x=0}^{1}\sum_{y=0}^{1} f\,(x,y)\left[\prod_{i=0}^{1-1}(-1)^{(b_i(x)b_{-i}(u)+b_i(y)b_{-i}(v))}\right]$$

$$H(u,v) = \frac{1}{N}\sum_{x=0}^{1}\sum_{y=0}^{1} f\,(x,y)(-1)^{(b_0(x)b_0(u)+b_0(y)b_0(v))}$$

$$= \frac{1}{2}f(0,0)(-1)^{(b_0(0)b_0(u)+b_0(0)b_0(v))} + \frac{1}{2}f(0,1)(-1)^{(b_0(0)b_0(u)+b_0(1)b_0(v))}$$
$$+ \frac{1}{2}f(1,0)(-1)^{(b_0(1)b_0(u)+b_0(0)b_0(v))} + \frac{1}{2}f(1,1)(-1)^{(b_0(1)b_0(u)+b_0(1)b_0(v))}$$
$$= \frac{1}{2}f(0,0)(-1)^{(0\cdot b_0(u)+0\cdot b_0(v))} + \frac{1}{2}f(0,1)(-1)^{(0\cdot b_0(u)+1\cdot b_0(v))}$$
$$+ \frac{1}{2}f(1,0)(-1)^{(1\cdot b_0(u)+0\cdot b_0(v))} + \frac{1}{2}f(1,1)(-1)^{(1\cdot b_0(u)+1\cdot b_0(v))}$$
$$= \frac{1}{2}f(0,0)(-1)^0 + \frac{1}{2}f(0,1)(-1)^{b_0(v)} + \frac{1}{2}f(1,0)(-1)^{b_0(u)}$$
$$+ \frac{1}{2}f(1,1)(-1)^{b_0(u)+b_0(v)}$$
$$= \frac{3}{2}(-1)^0 + \frac{1}{2}(-1)^{b_0(v)} + \frac{1}{2}(-1)^{b_0(u)} + \frac{2}{2}(-1)^{b_0(u)+b_0(v)}$$
$$= \frac{3}{2} + \frac{1}{2}(-1)^{b_0(v)} + \frac{1}{2}(-1)^{b_0(u)} + (-1)^{b_0(u)+b_0(v)}$$

$$H(0,0) = \frac{3}{2} + \frac{1}{2}(-1)^{b_0(0)} + \frac{1}{2}(-1)^{b_0(0)} + (-1)^{b_0(0)+b_0(0)}$$
$$= \frac{3}{2} + \frac{1}{2}(-1)^{0} + \frac{1}{2}(-1)^{0} + (-1)^{0} = \frac{3}{2} + \frac{1}{2} + \frac{1}{2} + 1 = \frac{7}{2}$$
$$H(0,1) = \frac{3}{2} + \frac{1}{2}(-1)^{b_0(0)} + \frac{1}{2}(-1)^{b_0(1)} + (-1)^{b_0(0)+b_0(1)}$$
$$= \frac{3}{2} + \frac{1}{2}(-1)^{0} + \frac{1}{2}(-1)^{1} + (-1)^{0+1} = \frac{3}{2} + \frac{1}{2} - \frac{1}{2} - 1 = \frac{1}{2}$$
$$H(1,0) = \frac{3}{2} + \frac{1}{2}(-1)^{b_0(1)} + \frac{1}{2}(-1)^{b_0(0)} + (-1)^{b_0(1)+b_0(0)}$$
$$= \frac{3}{2} + \frac{1}{2}(-1)^{1} + \frac{1}{2}(-1)^{0} + (-1)^{1+0} = \frac{3}{2} - \frac{1}{2} + \frac{1}{2} - 1 = \frac{1}{2}$$
$$H(1,1) = \frac{3}{2} + \frac{1}{2}(-1)^{b_0(1)} + \frac{1}{2}(-1)^{b_0(1)} + (-1)^{b_0(1)+b_0(1)}$$
$$= \frac{3}{2} + \frac{1}{2}(-1)^{1} + \frac{1}{2}(-1)^{1} + (-1)^{1+1} = \frac{3}{2} - \frac{1}{2} - \frac{1}{2} + 1 = \frac{3}{2}$$

Therefore, $H(u,v) = \begin{bmatrix} \frac{7}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{3}{2} \end{bmatrix}$.

(b) Consider the population of vectors $\underline{f}$ of the form

$$\underline{f}(x,y) = \begin{bmatrix} f_1(x,y) \\ f_2(x,y) \\ \vdots \\ f_N(x,y) \end{bmatrix}.$$

Each component $f_i(x,y),\ i = 1,2,\dots,N$ represents an image of size $M \times M$. The population arises from the formation of the vectors $\underline{f}$ across the entire collection of pixels $(x,y)$.

Consider now a population of vectors of the form:

$$\underline{g}(x,y) = \begin{bmatrix} g_1(x,y) \\ g_2(x,y) \\ \vdots \\ g_N(x,y) \end{bmatrix}$$

where the vectors $\underline{g}$ are the Karhunen-Loeve (KL) transforms of the vectors $\underline{f}$.

(i) Prove and explain the relationship between the covariance matrix of $\underline{g}(x,y)$ and the covariance matrix of $\underline{f}(x,y)$. What is the structure of the covariance matrix of $\underline{g}(x,y)$?

**[3]**

**Answer**

Using the relationships

$\underline{g} = \underline{A}(\underline{x} - \underline{m_x})$

$\underline{g}^T = [\underline{A}(\underline{x} - \underline{m_x})]^T = (\underline{x} - \underline{m_x})^T \underline{A}^T$

we get

$\underline{g}\underline{g}^T = \underline{A}(\underline{x} - \underline{m_x})(\underline{x} - \underline{m_x})^T \underline{A}^T \Rightarrow E\{\underline{g}\underline{g}^T\} = E\{\underline{A}(\underline{x} - \underline{m_x})(\underline{x} - \underline{m_x})^T \underline{A}^T\}$

$= \underline{A} E\{(\underline{x} - \underline{m_x})(\underline{x} - \underline{m_x})^T\} \underline{A}^T$

$\underline{C_g} = \underline{A}\, \underline{C_x} \underline{A}^T$

$\underline{C_x}\underline{A}^T = \underline{C_x}[\underline{e_1} \quad \underline{e_2} \quad \dots \quad \underline{e_n}] = [\lambda_1 \underline{e_1} \quad \lambda_2 \underline{e_2} \quad \dots \quad \lambda_n \underline{e_n}]$

$$\underline{C_g} = \underline{A}\underline{C_x}\underline{A}^T = \begin{bmatrix} \underline{e_1}^T \\ \underline{e_2}^T \\ \vdots \\ \underline{e_n}^T \end{bmatrix} [\lambda_1 \underline{e_1} \quad \lambda_2 \underline{e_2} \quad \dots \quad \lambda_n \underline{e_n}]$$

Because $\underline{e_i}$ is a set of orthonormal eigenvectors we have that:
$\underline{e_i}^T \underline{e_j} = 1, i = 1, \dots, n$
$\underline{e_i}^T \underline{e_j} = 0, i, j = 1, \dots, n$ and $i \neq j$
and therefore, $\underline{C_g}$ is a diagonal matrix whose elements along the diagonal are the eigenvalues of $\underline{C_x}$

(ii) Can the covariance matrix of $\underline{g}(x, y)$ be non-symmetric? Justify your answer.

**[2]**

**Answer**

The $(i, j)$ element of the covariance matrix of population $\underline{g}(x, y)$ is the covariance between images $i$ and $j$ and therefore, it is the same with the element $(j, i)$ of the covariance matrix of population $\underline{g}(x, y)$.

(iii) Suppose that $N = 8$ and that the eigenvalues of the covariance matrix of $\underline{f}(x, y)$ are:

$$[168 \quad 6.1 \quad 0.08 \quad 13 \quad 64 \quad 214 \quad 1.2 \quad 0.2]$$

What will be the mean square error if we use principal component images associated with the largest eigenvalues for 8: 1 and 4: 1 data compression? **[4]**

**Answer**

The mean square error for compression 8: 1 is $\sum_{j=2}^{8} \lambda_j = 252.58$.
The mean square error for compression 4: 1 is $\sum_{j=3}^{8} \lambda_j = 84.58$.

(c) Consider again Question 1(b) in the case of 3 images ($N = 3$). The covariance matrix of the population is $\underline{C_f}$ with elements defined as $C_{m,n} = 0.5^{|m-n|}, 1 \leq m, n \leq 3$.

(i) Write explicitly the covariance matrix $\underline{C_f}$ of the population of vectors $\underline{f}$. **[3]**

**Answer**

The covariance matrix of $\underline{f}$ is

$$\begin{bmatrix} 1 & 0.5 & 0.25 \\ 0.5 & 1 & 0.5 \\ 0.25 & 0.5 & 1 \end{bmatrix}$$

(ii) What will be the mean square error if we use the principal component image from the set of images $g_1(x, y)$, $g_2(x, y)$ and $g_3(x, y)$ to reconstruct the original images? The eigenvalues of matrix $\underline{C_f}$ are $\frac{1}{8}\left(9 + \sqrt{33}\right), \frac{3}{4}, \frac{1}{8}\left(9 - \sqrt{33}\right)$.

**[3]**

**Answer**

The error is $\frac{3}{4} + \frac{1}{8}\left(9 - \sqrt{33}\right)$.

2. (a) The histogram of a 3 bits/pixel square image is given in **Table 2.1** below.

| Grey level | Number of pixels |
|---|---|
| 0 | 4 |
| 1 | 4 |
| 2 | 0 |
| 3 | 0 |
| 4 | 0 |
| 5 | 3 |
| 6 | 2 |
| 7 | 3 |

**Table 2.1**

(i) Find out the size of image and the total number of pixels. **[2]**

**Answer**

$4 \times 4$ and 16 respectively.

(ii) What would be the new histogram if we make the MSB (Most Significant Bit) of all pixels equal to 1? **[3]**

**Answer**

| Original grey level | New grey level | Number of pixels |
|---|---|---|
| 0 | 4 | 4 |
| 1 | 5 | 4 |
| 2 | 6 | 0 |
| 3 | 7 | 0 |
| 4 | 4 | 0 |
| 5 | 5 | 3 |
| 6 | 6 | 2 |
| 7 | 7 | 3 |

| New grey level | Number of pixels |
|---|---|
| 4 | 4 |
| 5 | 7 |
| 6 | 2 |
| 7 | 3 |

(iii) Create a binary image with pixel values 0 and 7 using the technique of **Thresholding**. Compute the threshold $T$ such that $T = m$ where $m$ is the mean gray level value of the original image with histogram given in **Table 2.1** above. Apply this threshold and draw the new histogram for the output binary image. **[3]**

**Answer**

The mean of the original image is:

$$m = \frac{1 \times 4 + 5 \times 3 + 6 \times 2 + 7 \times 3}{16} = 3.25$$

| Original grey level | New grey level | Number of pixels |
|---|---|---|
| 0 | 0 | 4 |
| 1 | 0 | 4 |
| 2 | 0 | 0 |
| 3 | 0 | 0 |
| 4 | 7 | 0 |
| 5 | 7 | 3 |
| 6 | 7 | 2 |
| 7 | 7 | 3 |

| New grey level | Number of pixels |
|---|---|
| 0 | 8 |
| 7 | 8 |

(b) Which statistical parameters indicate the brightness of a grey level image among Standard Deviation, Mean and Median? Justify your answer by examining separately each of the given parameters. **[4]**

**Answer**

The Mean because it represents the average grey level value which demonstrates whether the image is dark or bright. The standard deviation demonstrates how high is the fluctuation of grey level values. An image can be constant with all the pixels have the maximum intensity and, therefore, very bright, and the variance is zero.

An image of size $M \times M$ with intensities between $0$ and $255$ might have $\frac{M \times M}{2} + 1$ intensities equal to $128$ and the rest equal to $255$. In that case the Median is $128$ which is the middle grey value. However, this is a bright image since the mean is:

$$\frac{1}{M \times M}\left[\left(\frac{M \times M}{2} + 1\right) \times 128 + \left(\frac{M \times M}{2} - 1\right) \times 255\right] \approx \frac{1}{2} \times 128 + \frac{1}{2} \times 255 \approx 192$$

(c) If the pixels of an image are rearranged which of the function/parameters below are affected? (i) Histogram, (ii) Mean, (iii) Median, (iv) Mode (v) None of the later. Justify your answer by examining separately each of the given parameters. **[4]**
[In statistics, the mode refers to the value that appears most frequently in a data set.]

**Answer**

None of the above, since all the paraneters given do not depend on the pixel locations but only on their collective values.

(d) What should the sum of all elements in an edge detection mask must be equal to? Justify your answer. **[4]**

**Answer**

The idea behind this is that the background pixels become $0$ after the application of the edge detection mask.

3. We are given the degraded version $g(x,y)$ of a digital image $f(x,y)$ such that in lexicographic ordering

$$g = Hf + n$$

where $H$ is the degradation matrix which is assumed to be block-circulant and $n$ is the noise term which is assumed to be zero-mean, white and independent of the image $f$. The symbols $f, n, g$ denote the lexicographically ordered original, noise and degraded image respectively. All images involved have size $N \times N$ after extension and zero-padding.

(a) (i) Provide the expression of the Inverse Filter in frequency domain that we may use to restore the original image. Under what assumptions can we perfectly restore the image using the technique of Inverse Filtering? **[3]**

**Answer**

In the presence of external noise we have that

$$\hat{F}(u,v) = \frac{H^*(u,v)\big(Y(u,v) - N(u,v)\big)}{\big|H(u,v)\big|^2} =$$

$$\frac{H^*(u,v)Y(u,v)}{\big|H(u,v)\big|^2} - \frac{H^*(u,v)N(u,v)}{\big|H(u,v)\big|^2} \Rightarrow$$

$$\hat{F}(u,v) = F(u,v) - \frac{N(u,v)}{H(u,v)}$$

If $H(u,v)$ becomes very small, the term $N(u,v)/H(u,v)$ dominates the result. Furthermore, if $H(u,v)$ is 0, the term $N(u,v)/H(u,v)$ is infinite. Therefore, the assumption which will enable us to perfectly restore the image is that there isn't any additive noise.

(ii) Explain why there might exist frequency pair(s) $(u,v)$ for which Inverse Filtering cannot be applied. **[3]**

**Answer**

This is because $H(u,v)$ is a lowpass filter and, therefore, depending on the particular type there are certain frequencies for which $H(u,v) = 0$. For example, if the spatial representation of the degradation model is a two-dimensional rectangular pulse, $H(u,v)$ is a *sinc* function which periodically becomes zero.

(iii) A particular image $f(x,y)$ is distorted by convolution with the space invariant point function $h(x,y)$ where:
$h(x,y) = \frac{1}{5}[\delta(x,y) + \delta(x-1,y) + \delta(x+1,y) + \delta(x,y-1) + \delta(x,y+1)]$
Show that for a specific value of $N$ there may exist at least one discrete frequency pair $(u,v)$ for which Inverse Filtering cannot be applied. Consequently, demonstrate how the method of Pseudoinverse Filtering for image restoration can be applied in this scenario. **[4]**

**Answer**

$$H(u,v) = \sum_{-1}^{1}\sum_{-1}^{1} h(x,y)e^{-j2\pi\left(\frac{xu+yv}{N}\right)}$$

$$= h(0,0) + h(0,1)e^{-j2\pi\frac{v}{N}} + h(0,-1)e^{j2\pi\frac{v}{N}} + h(1,0)e^{-j2\pi\frac{u}{N}} + h(-1,0)e^{j2\pi\frac{u}{N}}$$

$= 1 + e^{-j2\pi\frac{v}{N}} - e^{j2\pi\frac{v}{N}} + e^{-j2\pi\frac{u}{N}} - e^{j2\pi\frac{u}{N}} = 1 + 2\cos(2\pi\frac{u}{N}) + 2\cos\left(2\pi\frac{v}{N}\right)$

In the above relationship we can easily see that there are frequency pairs for which $H(u,v) = 0$, as for example $u = 2N/3$ and $v = N/4$ in case were both $2N/3$ and $N/4$ are integers. In that case

$$1 + 2\cos\left(2\pi\frac{2N}{3N}\right) + 2\cos\left(2\pi\frac{N}{4N}\right) = 1 + 2\cos\left(\frac{4\pi}{3}\right) + 2\cos\left(\frac{\pi}{2}\right)$$
$$= 1 + 2\left(\frac{-1}{2}\right) + 0 = 0$$

Therefore, Pseudo Inverse Filter must be used.

(b) The Constrained Least Squares (CLS) class of filters is designed to address potential issues that may arise during the application of Inverse Filtering to restore image $f$ from **Equation (3.1)** above. Their mathematical formulation is depicted below:

$$\frac{H(u,v)^*}{|H(u,v)|^2 + \alpha|C(u,v)|^2} \quad (\mathbf{3.2})$$

where $\alpha$ is the regularizations parameter and $C(u,v)$ represents the Laplacian operator. Show that there aren't any frequency pairs for which the original image cannot be recovered using **Equation (3.2)** above. **[4]**

**Answer**

The DFT of the Laplacian operator is given below:

$$C(u,v) = \sum_{-1}^{1}\sum_{-1}^{1} c(x,y)e^{-j2\pi\left(\frac{xu+yv}{N}\right)}$$
$$= c(0,0) + c(0,1)e^{-j2\pi\frac{v}{N}} + c(0,-1)e^{j2\pi\frac{v}{N}} + c(1,0)e^{-j2\pi\frac{u}{N}} + c(-1,0)e^{j2\pi\frac{u}{N}}$$
$$= 4 - e^{-j2\pi\frac{v}{N}} - e^{j2\pi\frac{v}{N}} - e^{-j2\pi\frac{u}{N}} - e^{j2\pi\frac{u}{N}} = 4 - 2\cos\left(2\pi\frac{u}{N}\right) - 2\cos\left(2\pi\frac{v}{N}\right)$$

We know that $0 \le u, v < N - 1$.
The only frequency pair for which $C(u,v) = 0$ is the pair $(u,v) = (0,0)$. However, for $(u,v) = (0,0)$ the term $|H(u,v)|^2$ is not zero because $H(u,v)$ is a lowpass filter. Therefore, there aren't any frequency pairs for which the original image cannot be recovered using equation above since

(c) Wiener Filtering is an alternative stochastic method used in image restoration to recover the image $f$ from **Equation (3.1)** above.

(i) Describe any type of statistical information regarding the original image and the additive noise that is required for implementing Wiener Filtering in image restoration. **[3]**

**Answer**

What is require is the power spectral density of both the noise and the original image.

(ii) Discuss potential strategies for obtaining the statistical information provided in (i) if it's not directly provided. **[3]**

**Answer**

The power spectral density is the Fourier transform of the autocorrelation function of the power signal. The autocorrelation function of white, zero mean noise is zero apart from the point 0, in the case of a two-dimensional noise function we can write:

$$R(k,l) = E\{n(x,y)n(x+k,y+l) = \sigma_n^2 \delta(k,l)$$

$\sigma_n^2$ is the variance of the noise which can be calculated from a flat area of the degraded image.
For the power spectral density of the original image, we can use the power spectral density of the degraded image. This is an approximation.

4. (a) A binary image is to be Huffman-coded in blocks of $M$ pixels. The successive pixel values are independent from each other. We assume that the pixel values are normalized, meaning that the maximum pixel value is 1. We know that 5% of the pixel values are 1 and the rest of the pixel values are 0. What are the Huffman codes for:

(i) $M = 1$ **[1]**

(ii) $M = 2$ **[2]**

(i) $M = 3$ **[3]**

(ii) For each of the above three cases, calculate the coding efficiency and the redundancy of each codebook.

**[2 points for each $M$; total 6 points]**

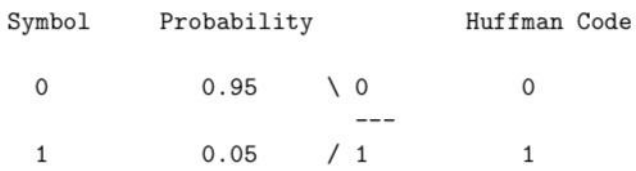
**Answer**

The minimum number of bits/symbol for lossless compression is given by the entropy

$$H = -\sum_i p_i \log_2(p_i)$$

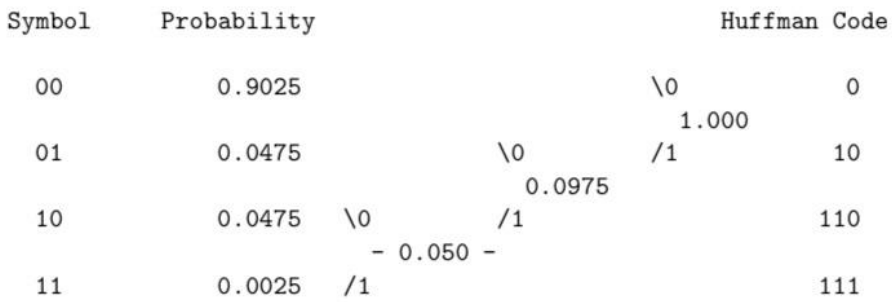
where $p_i$ is probability of each symbol.

(i) $M = 1$. In this case we code each pixel separately.



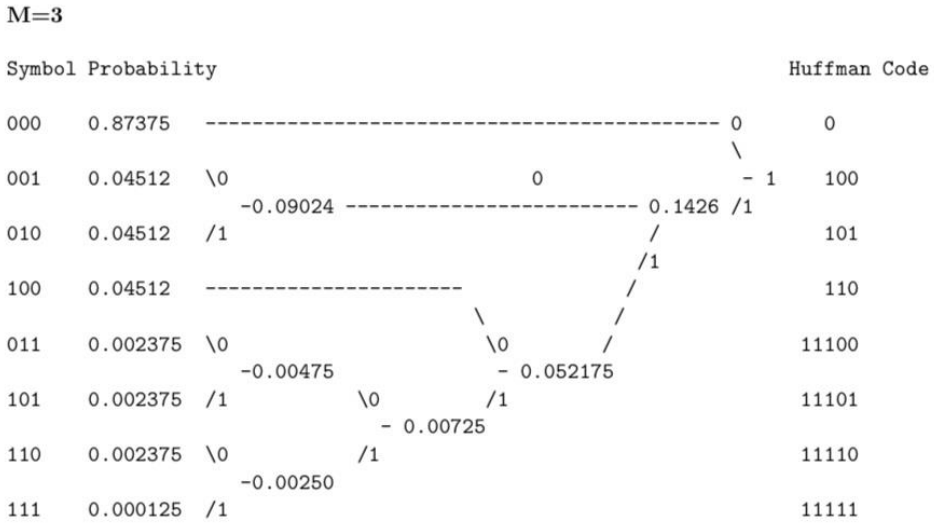

Entropy per symbol $H_1 = 0.286$ bits/pixel. Average length of Huffman code $= 1$. Efficiency is 0.28.

(ii) $M = 2$. In this case we encode pairs of pixels. Since we are told successive pixels are independent, we can easily calculate the probability of each possible combination of two pixels.




The entropy per symbol $H_2 = 0.572$ bits/symbol or 0.286 bits/pixel. Average length of the Huffman code is $1 \times 0.9025 + 2 \times 0.0475 + 3 \times 0.0475 + 3 \times 0.0025 = 1.147$. Efficiency $\frac{0.572}{1.147} = 0.5$.

(iii)




The entropy per symbol $H_3 = 0.859$ bits/symbol or $0.286$ bits/pixel. Average length of the Huffman code is 1.3bits. Efficiency $\frac{0.859}{1.3} = 0.66$.

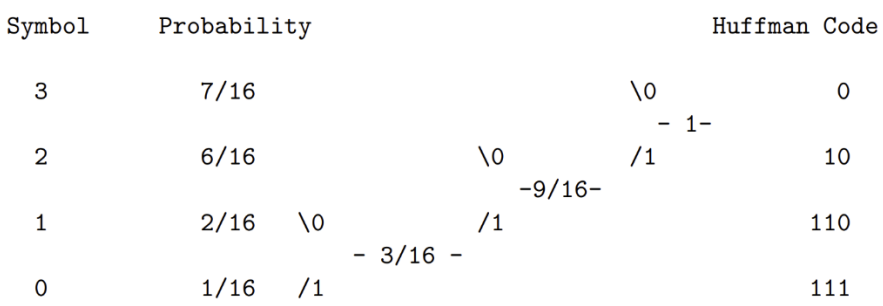
(b) A set of images is to be compressed. Each pixel of each image has a value in the range $[0, 3]$, i.e., we require 2 bits/pixel. An image taken from this set is given below. The frequency of occurrence of pixels of different values in this image, is typical of the set of images as a whole.

$$\begin{bmatrix} 3 & 3 & 3 & 2 \\ 2 & 3 & 3 & 3 \\ 3 & 2 & 2 & 2 \\ 2 & 1 & 1 & 0 \end{bmatrix}$$

What is the degree of compression achievable using the following methods? **[4]**

(i) Huffman coding of the pixel values.

**Answer**




The average number of bits is

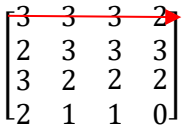
$$l_{\text{avg}} = \frac{2}{16} \cdot 3 + \frac{1}{16} \cdot 3 + \frac{7}{16} \cdot 1 + \frac{6}{16} \cdot 2 = 1.75$$

If we use fixed length symbol representation, we need 2 bits per symbol. Therefore, the compression ratio is $\frac{2}{1.75} = 1.1428$.

(ii) Forming differences between adjacent pixels (assuming horizontal row-by-row scan) and then Huffman coding these differences. For the top-left pixel use the value of 0 in this calculation. **[4]**

**Answer**

The zig-zag scanning shown below yields the matrix with 15 elements shown below as well.



$$\begin{bmatrix} 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & -1 & 0 & -1 \end{bmatrix}$$

| Symbol | Probability | Huffman Code |
|---|---|---|
| $-1$ | $\frac{4}{16}$ | 01 |
| 0 | $\frac{11}{16}$ | 1 |
| 1 | $\frac{1}{16}$ | 00 |

The average number of bits is $l_{\text{avg}} = \frac{4}{16} \cdot 2 + \frac{11}{16} \cdot 1 + \frac{1}{16} \cdot 2 = 1.25$

Compression ratio: $\frac{2}{1.25} = 1.6$